EXPERIMENT - 13

AIM :

To implement the Minimax algorithm for selecting the best move in a game tree.

ALGORITHM:

1. Start the program.
2. Define the leaf-node values of the game tree.
3. Evaluate the leaf nodes.
4. Select the maximum value at the MAX level.
5. Select the minimum value at the MIN level.
6. Continue until the root value is obtained.
7. Display the best value and stop.

PROGRAM :

untitled (3.14.8)

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```
def minimax(depth, node, is_max, values, max_depth):
    if depth == max_depth:
        return values[node]

    left = minimax(depth + 1, node * 2, not is_max,
                   values, max_depth)
    right = minimax(depth + 1, node * 2 + 1, not is_max,
                    values, max_depth)

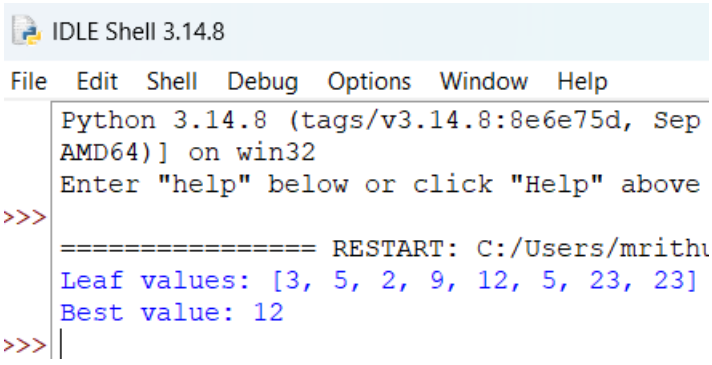
    if is_max:
        return max(left, right)
    else:
        return min(left, right)

values = [3, 5, 2, 9, 12, 5, 23, 23]

result = minimax(0, 0, True, values, 3)

print("Leaf values:", values)
print("Best value:", result)
```

OUTPUT:



RESULT:

Thus, the Minimax algorithm was implemented successfully to find the optimal value in a game tree